quiet-core conformance

Canonical anchors, derived roles, and contrast constraints

October 24, 2026

1 Purpose

This report records the canonical anchors and derived roles of quiet-day and quiet-night, and the evaluation of every quiet-core contrast constraint against them.

The tables are generated by `quiet report` from `spec/quiet-core.json`, `spec/quiet-day.json`, and `spec/quiet-night.json`. Constraints are evaluated by the same code as `quiet check`, so this report and the command cannot disagree.

The tables document what the reference implementation computes. Independent cross-checks of the underlying arithmetic live in the frozen test fixtures under `internal/`.

2 Method

Colors are sRGB with a D65 white point and standard dynamic range.

Contrast is the WCAG 2.x ratio of relative luminances, shown as Y.

Perceptual coordinates are OKLCH: lightness L, chroma C, and hue h in degrees. A hue is shown only when chroma is at least 0.020; below that, a numerically defined hue angle carries no meaning.

Ratios are printed to two decimals. Pass or fail is decided on unrounded values.

3 Canonical anchors

Table 1: quiet-day canonical anchors.

Anchor	Swatch	sRGB	L	C	h	Y
Field		#EEEADF	0.9371	0.0153	—	0.8235
Surface		#E6E9E2	0.9298	0.0099	—	0.8059
Structure		#E4EAED	0.9333	0.0076	—	0.8145
Ink		#1B1F1C	0.2340	0.0082	—	0.0130
Muted		#616B65	0.5173	0.0154	—	0.1400
Line		#CDD1CA	0.8556	0.0105	—	0.6284
Mineral		#4A6E7A	0.5157	0.0451	220.8	0.1401
Moss		#586E5C	0.5149	0.0384	150.2	0.1400
Ochre		#85602E	0.5171	0.0821	72.4	0.1355
Rust		#8F5A42	0.5207	0.0789	44.9	0.1355

4 Derived roles

Derived roles are reproduced from the anchors by the rules in quiet-core. Their values are consequences of those rules, not additional canonical colors.

Subtle is a compositing rule over the local background. It is listed here over Field for reference only.

5 Contrast constraints

Every constraint in the contract, with its minimum and the margin by which each adaptation exceeds it.

Table 2: quiet-night canonical anchors.

Anchor	Swatch	sRGB	L	C	h	Y
Field		#1C1E1C	0.2321	0.0048	—	0.0126
Surface		#242925	0.2745	0.0102	—	0.0209
Structure		#222A2D	0.2786	0.0125	—	0.0219
Ink		#E2E0D7	0.9058	0.0123	—	0.7439
Muted		#9AA29C	0.7040	0.0125	—	0.3511
Line		#343B38	0.3447	0.0108	—	0.0414
Mineral		#789AA3	0.6639	0.0399	215.7	0.2975
Moss		#879A83	0.6646	0.0398	140.2	0.2990
Ochre		#B49A67	0.6975	0.0749	83.6	0.3379
Rust		#C47D6B	0.6597	0.0931	34.9	0.2746

Table 3: quiet-day derived roles.

Role	Swatch	sRGB	Rule
Selection		#D4DDE1	OKLab mix, Structure toward Mineral
Callout		#E8ECE2	OKLCH adjustment of Surface
Subtle over Field		#A0A49C	sRGB composite of Muted over Field
Mineral wash		#E1EEE9	OKLab displacement of Field
Mineral wash emphasis		#C9E7E9	OKLab displacement of Field
Moss wash		#E5EED8	OKLab displacement of Field
Moss wash emphasis		#D3E8C7	OKLab displacement of Field
Ochre wash		#F7E8D1	OKLab displacement of Field
Ochre wash emphasis		#F5DCB8	OKLab displacement of Field
Rust wash		#FAE6D4	OKLab displacement of Field
Rust wash emphasis		#FCD8C0	OKLab displacement of Field

Table 4: quiet-night derived roles.

Role	Swatch	sRGB	Rule
Selection		#344246	OKLab mix, Structure toward Mineral
Callout		#262D28	OKLCH adjustment of Surface
Subtle over Field		#616762	sRGB composite of Muted over Field
Mineral wash		#102122	OKLab displacement of Field
Mineral wash emphasis		#042A30	OKLab displacement of Field
Moss wash		#172015	OKLab displacement of Field
Moss wash emphasis		#182A15	OKLab displacement of Field
Ochre wash		#211D11	OKLab displacement of Field
Ochre wash emphasis		#2D2309	OKLab displacement of Field
Rust wash		#251B16	OKLab displacement of Field
Rust wash emphasis		#341E15	OKLab displacement of Field

Table 5: Every quiet-core contrast constraint.

Constraint	Minimum	quiet-day	Margin	quiet-night	Margin
Ink on Field	7.00	13.87	+6.87	12.68	+5.68
Ink on Structure	7.00	13.73	+6.73	11.05	+4.05
Ink on Selection	4.50	12.08	+7.58	7.93	+3.43
Muted on Field	4.50	4.60	+0.10	6.41	+1.91
Muted on Surface	4.50	4.51	+0.01	5.65	+1.15
Muted on Structure	4.50	4.55	+0.05	5.58	+1.08
Muted on Callout	4.50	4.62	+0.12	5.39	+0.89
Mineral on Field	4.50	4.59	+0.09	5.55	+1.05
Mineral on Surface	4.50	4.50	+0.00	4.90	+0.40
Mineral on Structure	4.50	4.55	+0.05	4.84	+0.34
Moss on Field	4.50	4.60	+0.10	5.58	+1.08
Moss on Surface	4.50	4.51	+0.01	4.92	+0.42
Moss on Structure	4.50	4.55	+0.05	4.86	+0.36
Ochre on Field	4.50	4.71	+0.21	6.20	+1.70
Ochre on Surface	4.50	4.61	+0.11	5.47	+0.97
Ochre on Structure	4.50	4.66	+0.16	5.40	+0.90
Rust on Field	4.50	4.71	+0.21	5.19	+0.69
Rust on Surface	4.50	4.62	+0.12	4.58	+0.08
Rust on Structure	4.50	4.66	+0.16	4.52	+0.02
Ink on Mineral wash	7.00	13.95	+6.95	12.62	+5.62
Muted on Mineral wash	4.50	4.62	+0.12	6.38	+1.88
Mineral on Mineral wash	4.50	4.62	+0.12	5.52	+1.02
Moss on Mineral wash	4.50	4.62	+0.12	5.55	+1.05
Ochre on Mineral wash	4.50	4.73	+0.23	6.17	+1.67
Rust on Mineral wash	4.50	4.74	+0.24	5.16	+0.66
Ink on Mineral wash emphasis	7.00	12.79	+5.79	11.53	+4.53
Ink on Moss wash	7.00	13.97	+6.97	12.62	+5.62
Muted on Moss wash	4.50	4.63	+0.13	6.37	+1.87
Mineral on Moss wash	4.50	4.63	+0.13	5.52	+1.02
Moss on Moss wash	4.50	4.63	+0.13	5.55	+1.05
Ochre on Moss wash	4.50	4.74	+0.24	6.17	+1.67
Rust on Moss wash	4.50	4.74	+0.24	5.16	+0.66
Ink on Moss wash emphasis	7.00	12.83	+5.83	11.52	+4.52
Ink on Ochre wash	7.00	13.85	+6.85	12.69	+5.69
Muted on Ochre wash	4.50	4.59	+0.09	6.41	+1.91
Mineral on Ochre wash	4.50	4.59	+0.09	5.56	+1.06
Moss on Ochre wash	4.50	4.59	+0.09	5.58	+1.08
Ochre on Ochre wash	4.50	4.70	+0.20	6.20	+1.70
Rust on Ochre wash	4.50	4.70	+0.20	5.19	+0.69
Ink on Ochre wash emphasis	7.00	12.60	+5.60	11.68	+4.68
Ink on Rust wash	7.00	13.80	+6.80	12.75	+5.75
Muted on Rust wash	4.50	4.57	+0.07	6.44	+1.94
Mineral on Rust wash	4.50	4.57	+0.07	5.58	+1.08
Moss on Rust wash	4.50	4.57	+0.07	5.61	+1.11
Ochre on Rust wash	4.50	4.68	+0.18	6.23	+1.73
Rust on Rust wash	4.50	4.69	+0.19	5.21	+0.71
Ink on Rust wash emphasis	7.00	12.52	+5.52	11.80	+4.80

6 Tightest margins

The constraints with the least headroom across both adaptations, printed to three decimals so that a narrow pass does not read as a failure.

Table 6: Lowest margins across both adaptations.

Palette	Constraint	Ratio	Minimum	Margin
quiet-day	Mineral on Surface	4.502	4.50	+0.002
quiet-day	Moss on Surface	4.505	4.50	+0.005
quiet-day	Muted on Surface	4.506	4.50	+0.006
quiet-night	Rust on Structure	4.518	4.50	+0.018
quiet-day	Mineral on Structure	4.547	4.50	+0.047
quiet-day	Moss on Structure	4.550	4.50	+0.050
quiet-day	Muted on Structure	4.551	4.50	+0.051
quiet-day	Mineral on Rust wash	4.571	4.50	+0.071

quiet-day's chromatic and secondary foregrounds sit close to the 4.5:1 floor on Surface and Structure. This is deliberate: quiet-core places quiet-day's four chromatic foregrounds on a nearly common lightness and contrast plane.

The consequence is a constraint on change control. Any proposed change that lowers quiet-day Surface lightness, or lightens Mineral, Moss, or Muted, is likely to fail the contract unless it is compensated elsewhere.

quiet-night has more headroom. Its tightest relationship is Rust on Structure.

7 Reproduction

```
make report-data
make reports
quiet check spec/quiet-core.json spec/quiet-day.json spec/quiet-night.json
```

`make report-data` regenerates the tables. The test suite fails if the committed tables differ from current output. `make reports` exports this document to PDF.